Smart Kitchen Assistant

Week 7 Practice - Problem 4 (Intermediate)

KitchenTool encapsulates speedLevel in a private field and exposes JavaBean-style methods. The setter checks the range before assigning. Blender also implements Washable.

Java solution

```
abstract class KitchenTool {
    private int speedLevel = 1;
    public abstract String prepare();

    int getSpeedLevel() { return speedLevel; }

    void setSpeedLevel(int speedLevel) {
        if (speedLevel < 1 || speedLevel > 5) {
            throw new IllegalArgumentException(
                "speedLevel must be between 1 and 5");
        }
        this.speedLevel = speedLevel;
    }
}

interface Washable {
    String clean();
}

class Blender extends KitchenTool implements Washable {
    public String prepare() {
        return "Blending at speed " + getSpeedLevel();
    }
    public String clean() { return "Blender rinsed and dried"; }
}

public class Main {
    public static void main(String[] args) {
        Blender b = new Blender();
        b.setSpeedLevel(3);
        System.out.println(b.getSpeedLevel());
        try {
            b.setSpeedLevel(9);
        } catch (IllegalArgumentException e) {
            System.out.println("rejected, speed level stays "
                + b.getSpeedLevel());
        }
        System.out.println(b.prepare());
        System.out.println(b.clean());
    }
}
```

Sample output

```
3
rejected, speed level stays 3
Blending at speed 3
Blender rinsed and dried
```

Note: Invalid values throw IllegalArgumentException and leave the previously accepted speed unchanged.